

SUDIP DAS

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Senior AI ML Engineer | IIT Gn | 3.5+ years building scalable LLM & GenAI systems

Summary: Experienced in developing domain-adapted LLM pipelines, RAG systems, and OCR+CV workflows for legal, financial, and IP domains. Proven track record in cost-saving deployments, model evaluation and finetuning. Seeking impactful Generative AI ML roles where research meets production.

EDUCATION

M.Tech in Computer Science

IIT Gandhinagar, *GPA: 8.44*

Gandhinagar, Gujarat

August 2020 - August 2022

B.Tech in Computer Science

Techno Main Salt Lake, *GPA: 8.35*

Salt Lake, West Bengal

August 2016 - August 2020

CORE SKILLS

Programming & Scripting: Python (Advanced), Jupyter, Git, FastAPI

AI ML Frameworks: PyTorch, vLLM, Ollama, Hugging Face Transformers

NLP: Named Entity Recognition, Summarization, Guardrails, Evaluation Pipelines

ADVANCED SKILLS

Generative AI: Retrieval Augmented Generation, Chain-of-Thought & ReAct Prompting

LLM Agents: Langchain, LangGraph, LangSmith, Agent2Agents, Model Context protocol

LLM Core Development: LLM Pretraining & Supervised Finetuning, Model Evaluation, QLORA, PEFT

Computer Vision: Layout Parser, Object Segmentation, Vision Transformers, OpenCV, OCR

Database & Messaging Queues: PostgreSQL, MySQL, MongoDB, ElasticSearch, Qdrant, RabbitMQ

Cloud & MLOps: Docker, Kubernetes, Azure AI Foundry, AWS, S3, SQS, EC2, Bedrock, Sage-maker, CI/CD, MLflow, Model Deployment, Monitoring

INDUSTRY EXPERIENCE

smartSense consulting solutions Pvt. Ltd.

Associate AI ML Engineer → AI ML Engineer → Senior AI ML Engineer

Gandhinagar, India

June 2022 – Present

Senior AI ML Engineer

April 2025 – Present

- **AI-Powered Legal Document Automation Platform (Outsourced.co, NZ LegalTech SaaS)**

Tech Stack: Python, FastAPI, MongoDB, Milvus, Azure AI Foundry

- Developed and deployed an **LLM-driven platform** automating the generation of affidavits, statements of claim, and legal notices, streamlining compliance for New Zealand legal workflows.
- Designed and implemented a robust **preprocessing pipeline** to extract data from PDFs, HTML, and other formats, feeding structured information into a reasoning model for advanced context understanding.
- Engineered and productionized a **modular prompt templating system** leveraging structured assets (guides, examples, templates, guardrails) to dynamically assemble context-rich prompts for the LLM, ensuring high-quality and legally compliant document generation.
- Conducted extensive R&D and benchmarking to select the optimal reasoning model, ultimately choosing **O4 Mini** for its superior performance in legal document automation.
- Implemented a **Milvus-powered RAG workflow** for semantic invoice search, enabling rapid retrieval and contextualization of legal and financial documents.

AI ML Engineer

July 2023 – April 2025

- **Graph-Based AI Agents (Second Brain, Internal B2B Product)**

Tech Stack: LangGraph, Langchain, FastAPI, Celery, Azure ML Studio, ElasticSearch, MongoDB, Qdrant, RAGAS

- Architected and implemented Leave and Onboarding Agents using LangGraph, orchestrating multi-agent workflows based on user queries.
- Designed an AI-powered knowledge management system that organizes departmental documents into secure vectorized knowledge bases, also known as brains.

- **Generative AI-Powered PDF2HTML for Legal Documents (New Zealand Legal Firm)**

Tech Stack: EC2, Detectron2, Anthropic Sonnet 3.5, Gemini 1.5 Flash

- Developed a GPU-accelerated, on-premise OCR pipeline using Layout Parser to convert **100K+ legal documents** from PDF to structured HTML, enabling downstream automation and analytics.
- Leveraged **Gemini 1.5 Flash** to further process extracted tables of contents into precise HTML code, and analyzed images to generate rich metadata for legal workflows.
- Achieved significant operational efficiency and delivered **\$400K+ in cost savings** by replacing proprietary cloud solutions with an optimized on-premise pipeline.

- **Patent and Image Drafting Automation (Client: One of the World's Largest Chemical Companies, Germany)**

Tech Stack: vLLM, QLoRA, Finetuning, Pretraining, Prompt Engineering, Model Benchmarking

- Led a research initiative to enhance patent drafting efficiency by finetuning open-source LLMs (Falcon, LLaMA 2, Mistral, Mixtral) for legal text generation.
- Increased context window of LLMs with SOTA methods (YARN, RoPE), enabling concise and effective patent drafts and boosting attorney productivity by **30%**.
- Designed an LLM-powered system to generate accurate SVG diagrams from text, achieving **94% accuracy** and automating image drafting.

Associate AI ML Engineer

June 2022 – July 2023

- **Data Science Platform for Banks, Professionals & Remote Nodes (Bangalore Data Science Startup)**

Tech Stack: FastAPI, PostgreSQL, MongoDB, TensorFlow, PyTorch

- Led a team of 5 to develop a distributed ML training platform, connecting banks, data scientists, and remote nodes.
- Conducted hackathons with non-sensitive datasets, selecting top talents for banks' AI initiatives.
- Implemented secure dataset access using a private blockchain-based validation.

- **IntelliBot - Scalable Conversational AI Bot (Client: One of the World's Largest Chemical Companies, Germany)**

Tech Stack: RASA, Hugging Face, MongoDB

- Built a transformer-based chatbot handling **50+ intents** with secure departmental access controls.
- Developed an ML-driven spell checker integrated with NER models, efficiently managing **2000+ entities**.

ACHIEVEMENTS & CERTIFICATIONS

AWS Certified AI Practitioner

Valid till July 2028

All India Rank 8 JEST 2020

IMSc

Also cleared Round 2

March 2020

All India Rank 616 GATE 2020

IIT Delhi

With a percentile of 99.4%

April 2020

NPTEL Course DBMS

IIT Kharagpur

Gold Medalist

April 2019

PUBLICATIONS

- Athulkumar R, Aarchi Agrawal, K S Ashin Shanly, **Sudip Das** and Nidhin Harilal
[Image Caption Generator using Siamese Graph Convolutional Networks and LSTM](#)
ACM IKDD CODS 2022
- Vridhi Rohira, Shivam Chaudhary, **Sudip Das**, Krishna Prasad Miyapuram
[Automatic Epilepsy Detection from EEG signals](#)
ACM IKDD CODS 2023
- **Sudip Das**, Pankaj Pandey, Krishna Prasad Miyapuram
[Improving self-supervised pretraining models for epileptic seizure detection from EEG data](#)
Master Thesis project (Pre-print)